Self-Reflective Purchase Intention Agent

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An open-source, local-first system for explainable customer purchase-intention prediction, self-reflection, and authenticated deployment.

Executive Summary

This project implements an end-to-end purchase-intention agent without paid APIs or proprietary services. It generates privacy-safe behavioural data for reproducible experimentation, trains a calibrated logistic-regression model, exposes signed feature contributions and counterfactual guidance, and runs a deterministic self-reflection pass before returning a decision.

The system is available locally through FastAPI and a browser workspace, and is deployed to Vercel through a lightweight pure-Python inference adapter. Credential-based access protects hosted prediction routes with an HTTP-only signed session cookie.

Project Objectives

1. Predict whether a customer session is likely to result in a purchase. 2. Make each prediction understandable through signed behavioural factors. 3. Detect low-confidence and boundary cases that deserve human review. 4. Keep the stack free, open source, reproducible, and deployable without secrets in source control.

System Architecture

The local research path uses Python, pandas, NumPy, scikit-learn, FastAPI, SQLite, and vanilla HTML/CSS/JavaScript. The Vercel path uses FastAPI, Pydantic, and standard-library Python only, avoiding the serverless function size limit caused by scientific dependencies.

Layer	Responsibility	Implementation
Data	Generate and validate behavioural aggregates	data.py
Model	Train and persist calibrated classifier	model.py
Explain	Expose signed local feature impacts	explain.py
Reflect	Assess confidence, quality, and stability	reflection.py
Serve	API, auth, UI, and audit records	api.py / api/index.py
Deploy	Public production hosting	Vercel + vercel.json

Data and Features

The reproducible generator creates session-duration, pages-viewed, product-views, cart-additions, prior-purchases, discount usage, device type, traffic source, and returning-customer features. The target is a purchase observed within the chosen session window. The validation layer rejects missing, non-finite,

out-of-range, or internally inconsistent inputs.

Model and Evaluation

The baseline model is a logistic regression pipeline with standardised numeric features and one-hot encoded categorical features. On the verified 800-sample training run with seed 11, the held-out results were:

Metric	Value	Interpretation
ROC-AUC	0.7949	Useful ranking discrimination
Accuracy	0.7375	Correct class decisions at 0.5
Brier score	0.1801	Probability calibration error
Test samples	160	Held-out evaluation size

Explainability and Self-Reflection

For each scored session, the agent returns positive and negative feature contributions derived from the transformed model input and learned coefficients. It also returns a risk band, a human-readable summary, and a counterfactual suggestion such as increasing high-intent engagement.

The reflection pass checks four conditions: behavioural input quality, confidence away from the decision boundary, explanation availability, and boundary stability. Low-confidence or invalid cases are marked for human review rather than presented as certain conclusions.

Authentication and Privacy

The hosted app presents a credential login interface. Username, password, and session secret are read from environment variables. Successful login creates an HTTP-only, signed, eight-hour cookie; prediction and metrics endpoints reject unauthenticated requests. The repository does not contain passwords or session secrets.

The demo uses behavioural aggregates rather than names, emails, addresses, or raw event payloads. Hosted SQLite audit data is written to ephemeral /tmp storage and should be replaced with a durable, access-controlled database for production use.

Deployment

Production URL: https://purchase-intention-agent.vercel.app. Vercel hosts the lightweight adapter from api/index.py, while local development retains the full scikit-learn research path. The deployment is configured by vercel.json and requirements.txt.

Validation and Reproducibility

The repository test suite passes 3 tests covering model discrimination, explanation/reflection output, input validation, and the local API prediction contract. Additional smoke checks verified the Vercel dashboard, health route, login flow, unauthorized 401 response, and authenticated prediction response.

API Surface

Route	Purpose	Access
GET /	Authenticated workspace shell	Public shell
POST /login	Create signed session	Public
POST /logout	Clear session	Public
GET /health	Service status	Public
GET /model/metrics	Model validation metrics	Authenticated
POST /predict	Score and explain a session	Authenticated

Limitations and Next Steps

The current dataset is synthetic and should not be used to make business decisions without retraining and validating on consented production data. Recommended extensions are temporal and group-aware splits,

subgroup calibration, drift monitoring, durable audit storage, human feedback capture, and an optional local Ollama or vLLM explanation writer behind the deterministic contract.

Generated from the open-source project repository. No proprietary model or paid API is required.